

Machine Learning in Finance

Deconstructing published papers

This is not a methods survey. It is a **research-design** workshop.

Barbara Będowska-Sójka · Poznań University of Economics and Business

MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



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Learning goals

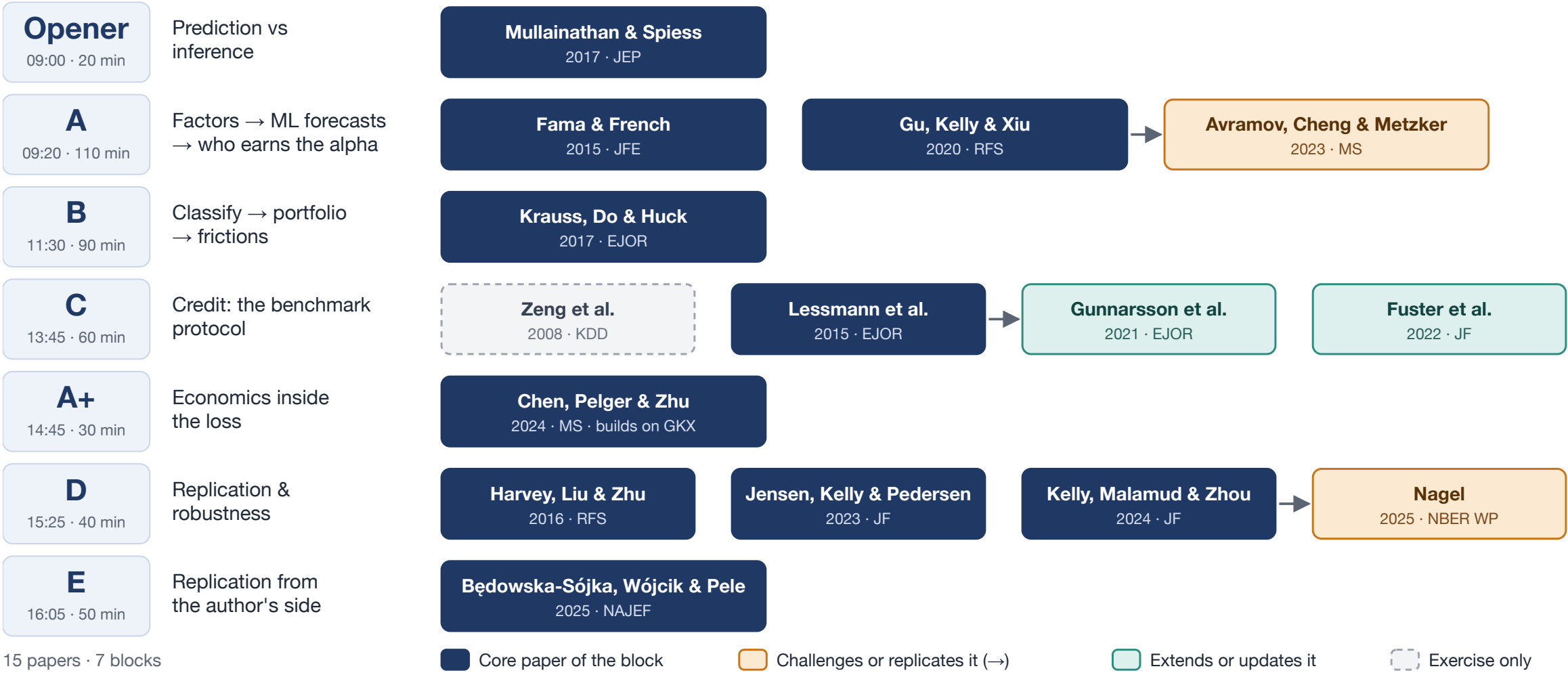
By the end of today you can:

1. Restate a paper as an **economic question + prediction design**
2. Tell a **prediction** apart from a **hypothesis test**
3. Spot **leakage**, bad validation and metric mismatch
4. Walk a pipeline: sample → features → split → models → finance evaluation
5. Judge a **robustness** or **replication** claim
6. Propose one falsifiable "**do differently**" experiment

Day map

Block	Focus	Min	Live artifact
Opener	pitfalls · R vs Python · prediction vs inference	20	this deck · 00b
A · FF → Gu → robustness	factors, ML forecast, who earns the alpha	110	01 , 01a
B · Krauss–Do–Huck	classify → portfolio → frictions	90	02
C · Credit	Lessmann protocol on public data · Fuster	60	03_lessmann_credit_lab
A+ · Chen–Pelger–Zhu	economics inside the loss	30	slides
D · Replication	what counts · KMZ vs Nagel live	40	06_kmz_nagel
E · Crypto zombies	replication from the author's side	50	05
07 · Cakici	ML and the crypto cross-section	—	07
Wrap	seven moves	5	08_wrap

The day's papers at a glance



Papers (1/2): asset pricing

Block	Citation
Opener	Mullainathan & Spiess (2017) <i>JEP, Machine Learning: An Applied Econometric Approach</i>
A	Fama & French (1993, 2015) <i>JFE, Common risk factors... / A five-factor asset pricing model</i>
A	Gu, Kelly & Xiu (2020) <i>RFS, Empirical Asset Pricing via Machine Learning</i>
A	Avramov, Cheng & Metzker (2023) <i>MS, Machine Learning vs. Economic Restrictions</i>
B	Krauss, Do & Huck (2017) <i>EJOR, ...Statistical arbitrage on the S&P 500</i>
A+	Chen, Pelger & Zhu (2024) <i>MS, Deep Learning in Asset Pricing</i>

Papers (2/2): credit, replication, crypto

Block	Citation
C	Lessmann, Baesens, Seow & Thomas (2015) <i>EJOR, Benchmarking ... credit scoring</i>
C	Gunnarsson et al. (2021) <i>EJOR, Deep learning for credit scoring: Do or don't?</i>
C	Fuster, Goldsmith-Pinkham, Ramadorai & Walther (2022) <i>JF, Predictably Unequal?</i>
D	Harvey, Liu & Zhu (2016) <i>RFS</i> ; Jensen, Kelly & Pedersen (2023) <i>JF</i>
D	Kelly, Malamud & Zhou (2024) <i>JF</i> vs Nagel (2025) NBER WP 34104
E	Będowska-Sójka, Wójcik & Pele (2025) <i>NAJEF, ...Predicting 'zombie' assets</i>

Used as an exercise only: Zeng, Melville et al. (2008) *KDD, invoice-to-cash collection*.

How we work each paper

We use the same 8 prompts every time (`sessions/SESSION_TEMPLATE.md`):

1. Economic question, and the **hypothesis**
2. Target & sample
3. Features
4. Validation
5. Model ladder
6. Finance evaluation
7. Replicability: data **and code**
8. Do differently

Paper card: Mullainathan & Spiess (2017)

	<i>Journal of Economic Perspectives</i> 31(2): 87–106
Question	What is machine learning good for in applied economics, and what is it not good for?
Data	American Housing Survey 2011: predict house values from 150 variables; 10,000 training and 41,808 hold-out units.
Method	OLS vs regression tree, LASSO, random forest and an ensemble, tuned by cross-validation.
How it is tested	Hold-out R^2 ; LASSO re-fitted on ten random partitions of the data.
Main result	ML predicts better, but modestly: hold-out R^2 41.7% (OLS) vs 45.9% (ensemble). LASSO picks different variables in each partition with the same fit.
Watch for	"ML belongs in the $\hat{y}$ compartment, not the $\hat{\beta}$ compartment." A selected variable is not a finding.
Code & data	Replication files on openICPSR (project 113993).

Prediction vs inference

Mullainathan, S., & Spiess, J. (2017). *Machine Learning: An Applied Econometric Approach*. *Journal of Economic Perspectives*, 31(2), 87–106.

"Machine learning belongs in the part of the toolbox marked $\hat{y}$ rather than in the more familiar $\hat{\beta}$ compartment."

ML algorithms are built to predict. Even when they report coefficients, "the estimates are rarely consistent".

ML predicts well, but modestly

American Housing Survey 2011: 10,000 training units, 150 variables, 41,808 hold-out units.

Model	Hold-out R^2
OLS	41.7%
Regression tree	34.5%
LASSO	43.3%
Random forest	45.5%
Ensemble	45.9%

...but the selected variables are unstable

- LASSO re-fit on **ten partitions** of about 5,000 units each
- A different set of variables in each partition, with the same fit
- "Similar predictions can be produced using very different variables"

So a selected variable is not a finding. The same holds for feature importance (Gu, Block A) and XAI (crypto zombies, Block E).

Where ML helps research: new data (text, images) · first-stage prediction (IV, propensity scores) · prediction-policy problems.

Map the day: $\hat{y}$ or $\hat{\beta}$?

Paper	Prediction target	Hypothesis, if any
FF (2015)	none	Five factors price average returns ($\alpha = 0$)
Gu (2020)	next-month return	Nonlinear interactions carry risk-premium information
Krauss (2017)	$P(r > \text{CS median})$	Past prices predict returns (weak form)
Lessmann (2015)	$P(\text{default})$	Advanced classifiers beat LR under a fixed protocol
Fuster (2022)	$P(\text{default})$	ML changes who gains from credit
CPZ (2024)	SDF weights	No-arbitrage structure improves pricing
KMZ vs Nagel	market return	More parameters improve timing
Zombies (2025)	$P(\text{delisting, 28d})$	Zombie status is predictable from market data

We come back to this table in Block D.

Why finance breaks naive ML

- **Low SNR:** a tiny R^2 can still matter in portfolios
- **Time:** the future must not train the past
- **Decisions:** AUC $\neq$ Sharpe $\neq$ expected loss
- **Frictions:** turnover, shorting, collection capacity
- **Multiple testing:** the factor zoo is real

Pitfall 1: Look-ahead / leakage

Classic killers:

- Using revised filings as if they were known in real time
- Features updated *after* the label date
- Random CV on panels / invoices / customers
- Index membership known only ex post (survivors)
- LLMs trained on text from *after* the forecast date

Rule: write down the information set at decision time.

Pitfall 2: Wrong split

OK under i.i.d.	Dangerous in finance
Shuffle + K-fold	Leaks adjacent times / firms
Tune on test	Inflates horse-race winners
One lucky seed	Unstable rankings

Prefer a **calendar** train / validation / test split (expanding or rolling).

Pitfall 3: Metric mismatch

Domain	Statistical score	Economic object
Asset pricing	OOS R^2 ; GRS / mean $ \alpha $	long–short, Sharpe, costs
Stat-arb	AUC / accuracy	ranked portfolio + turnover
Credit	AUC / H / Brier	expected cost, calibration, who gains

Always ask: *what decision does the score feed?*

The model ladder (classroom habit)

Always keep a **strong simple baseline**.

linear / logit → regularized → trees / boosting → small NN

Complexity must buy something on the **economic** metric, not only on fit.

What the classroom notebooks are

Module	Data	Goal
FF (01)	real, 50 current names	attribution vs forecast; survivors
Gu (01a)	synthetic characteristics	pipeline, not a CRSP replica
Krauss (02)	synthetic, or real 50 names	label → portfolio → costs
Credit (03)	real, public (UCI; Kaggle if you download it)	a mini-Lessmann replication
KMZ–Nagel (06)	real, public (Welch–Goyal)	replicate, then reinterpret
Zombies (05)	real, public CSV	an author-led pipeline

With synthetic data the magnitudes will not match the paper. The **design choices** should.

Your course folder

- `00_START_HERE.pdf` : setup check and data sources
- `Programme.pdf` · `Reading_list.pdf` · `Paper_cards.pdf`
- `notebooks/` : the Python notebooks we run today
- `data/` : the data, with precomputed results in `data/cache/`
- `scripts/` : the scripts that precompute the slow steps
- The slides are in `slides/`, one deck per block

Ground rules for discussion

- Critique the **design**, not the authors' intelligence
- Prefer "how would we test that?" over vibes
- Park deep architecture rabbit holes for later
- One takeaway sentence per module

Next up: Module A

First Fama–French on 50 names (01), **then** Gu, **then** who earns Gu's alpha.

→ `slides/01_ff_factors.md` · `notebooks/01_ff_factors_real.ipynb`

→ `slides/01a_gu_kelly_xiu.md` · `notebooks/01a_gu_kelly_xiu_simplified.ipynb`